Water and Ion Partitioning, Confined Structure, and Axial Diffusion in Rigid Armchair Carbon Nanotubes

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1 Introduction

This report examines how the size of rigid armchair carbon nanotubes (CNTs) and the NaCl concentration of the surrounding reservoirs affect confined water and ions. The systems comprise eight CNT sizes with chiral indices (i, i), where $i = 6, 8, 9, 10, 12, 22, 30$, and 40, together with pure-water and six NaCl solution conditions. Each system was simulated for 500 ns using molecular dynamics in the NPT ensemble.

The analysis combines water and ion partitioning, axial and radial density profiles, static structure factors $S(q)$, and residence-conditioned axial self-diffusion estimates. Partitioning compares the central CNT region with a measured reservoir reference, while normalized diffusion ratios compare axial mobility inside the CNT with the corresponding reservoir or bulk reference. These observables are analyzed together because confinement changes composition, spatial organization, and mobility at the same time.

The report first summarizes the simulation systems and analysis procedures, then presents pure-water and saltwater results separately. The final sections compare concentration and diffusion ratios, summarize the confined structural regimes, and discuss the limitations of the current sampling and model. Detailed supplementary data and complete system scans are provided in the appendices.

2 Simulation Systems and Methods

2.1 Simulation systems

The study considers eight rigid armchair CNT sizes with chiral indices (i, i), where $i = 6, 8, 9, 10, 12, 22, 30$, and 40. Each CNT is aligned with z and connects two reservoirs (Figure 1), with periodic boundaries in all three directions. For each size, one pure-water system and six NaCl solution systems were simulated, giving 56 systems in total. The nominal NaCl concentrations were 0.2, 1, 2, 3, 4, and 5 M. The measured reservoir concentrations used for partitioning were obtained from the trajectories.

The axial span of the CNT carbon centers in the energy-minimized coordinates is approximately 6.026 nm. The geometric radius R is the mean radial distance of the tube carbon centers from the axis and is distinct from the fluid-accessible radius. For $i = 6, 8, 9, 10, 12, 22, 30$, and 40, the corresponding geometric radii are approximately 0.407, 0.542, 0.610, 0.678, 0.814, 1.492, 2.034, and 2.712 nm, respectively.

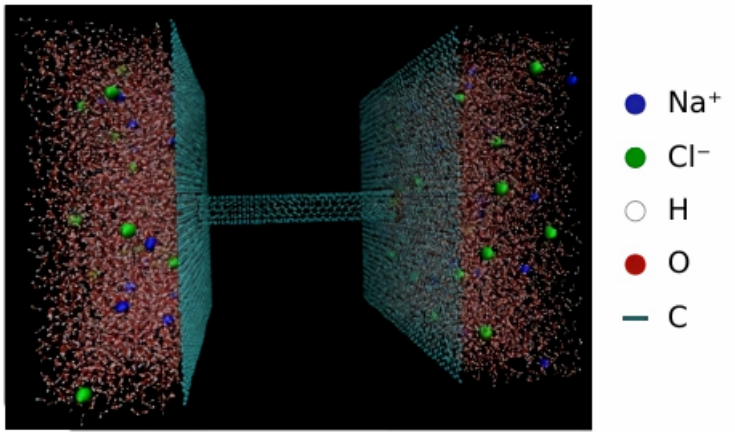


Figure 1: Representative simulation configuration for an open-ended rigid armchair CNT connecting two solution reservoirs. Pure-water systems use the same CNT–reservoir geometry without dissolved Na^+ and Cl^- ions.

2.2 Molecular dynamics protocol

Following steepest-descent minimization, each system was run for 20 ns with pressure coupling along z, then continued from a checkpoint for 480 ns without regenerating velocities. The combined NPT trajectory spans 500 ns per system. All NPT simulations were performed using GROMACS 2025.1. Table 1 summarizes the MD settings.

Table 1: Molecular dynamics settings.

Setting	Value
Water and ions	Rigid SPC/E (SETTLE), $q_{\mathrm{O}} = -0.8476e$, $q_{\mathrm{H}} = +0.4238e$, $q_{\mathrm{Na}} = +e$, $q_{\mathrm{Cl}} = -e$. O–H and H–H distances are 0.100 and 0.16330 nm.
Carbon and time step	Uncharged HCS carbon atoms frozen in x, y, z. Leap-frog integration, 2 fs time step.
Temperature	Velocity-rescale at 300 K, $\tau_T = 0.5$ ps. The input specifies separate coupling groups for the frozen carbon framework and the fluid.
Pressure	Semi-isotropic C-rescale, $\tau_P = 2.0$ ps. Fixed lateral dimensions, with pressure coupled to 1 bar along z. Compressibility is 0 in xy and $4.5 \times 10^{-5}\,\mathrm{bar}^{-1}$ along z.
Nonbonded interactions	Verlet neighbor scheme. PME with 0.9 nm real-space cutoff, 0.1 nm Fourier spacing and order 4 interpolation. LJ cutoff 0.9 nm, Potential-shift-Verlet, no long-range dispersion correction. See Appendix A.1 for LJ parameters.
Coordinate output	Compressed coordinates saved every 1000 steps (2 ps).

2.3 Analysis overview

Concentration and partitioning are evaluated over 300–500 ns using a centered 4.0 nm CNT region and a remote 2.0 nm reservoir slab. Reservoir concentrations are measured directly from the trajectories and provide the reference for partitioning. For species s, the partition coefficient is

$$K_s = \frac{c_{s,\mathrm{CNT}}}{c_{s,\mathrm{bulk}}}. \tag{1}$$

where s denotes the species (water, Na^+, or Cl^-), CNT denotes the central nanotube region, and bulk denotes the measured reservoir reference region.

Axial diffusion is estimated from the water-oxygen and individual-ion coordinates using

$$\mathrm{MSD}_z(t) = \left\langle [z_i(t_0 + t) - z_i(t_0)]^2 \right\rangle_{i,t_0}. \tag{2}$$

Here, i indexes particles and t_0 denotes the time origin. The average is taken over particles and time origins. Only particles that remain continuously inside the CNT over the interval from t_0 to $t_0 + t$ contribute. A linear fit of the mean-square displacement is performed according to

$$\mathrm{MSD}_z(t) = 2D_z t + b. \tag{3}$$

over $t = 20$–200 ps is used to estimate D_z. The intercept b is a fitted statistical offset; the diffusion coefficient is determined by the slope. The 0–500 ns time series uses 50 ns trajectory blocks, whereas the 300–500 ns ratio analysis uses 20 ns time-origin blocks. Thus, t is the elapsed MSD interval, while the block coordinates refer to positions within the full simulation trajectory. Axial and radial density profiles describe spatial organization, while $S(q)$ probes ordering in confined water. Block estimates may be correlated.

3 Pure-Water Results

Water forms a near-wall layer about 0.35 nm inward from the carbon wall, and the wider tubes approach the reservoir concentration toward the axis (Figure 2). The surface-normalized deficit length is lowest for $(8, 8)$ and varies little among the three widest tubes (Figure 3). Axial diffusion estimates cluster near $2.3 \times 10^{-9}\,\mathrm{m^2\,s^{-1}}$ from $(10, 10)$ onward, with lower values in narrower tubes (Figure 4).

3.1 Radial Water Distribution

Here, $r = \left[(x - x_c)^2 + (y - y_c)^2\right]^{1/2}$ is the particle distance from the CNT axis, where (x_c, y_c) denotes the axis centre. R is the mean carbon-wall-centre radius, so $R - r$ measures the inward distance from the wall-centre radius toward the CNT axis.

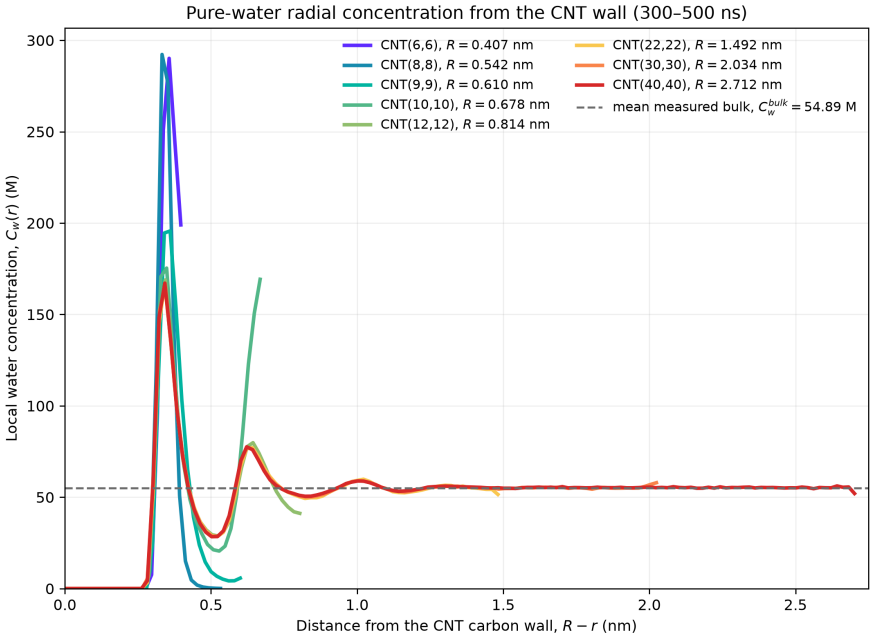


Figure 2: Local water concentration versus wall distance, 300–500 ns. The dashed line marks the reservoir mean (54.89 M).

3.2 Surface-Normalized Deficit Length

For larger CNTs, δ_θ becomes nearly constant at about 0.18–0.19 nm. The smaller CNTs show stronger non-monotonic behavior, with CNT $(8, 8)$ giving the smallest value and therefore the most distinct confinement effect.

This figure shows the surface-normalized water deficit length (δ_θ) as a function of CNT radius during 300–500 ns. The concentration ratio is first calculated as $K_C = C_w^{\mathrm{CNT}}/C_w^{\mathrm{bulk}}$, and then converted to

$$\delta_\theta = \frac{R}{2}\left(1 - K_C\right). \tag{4}$$

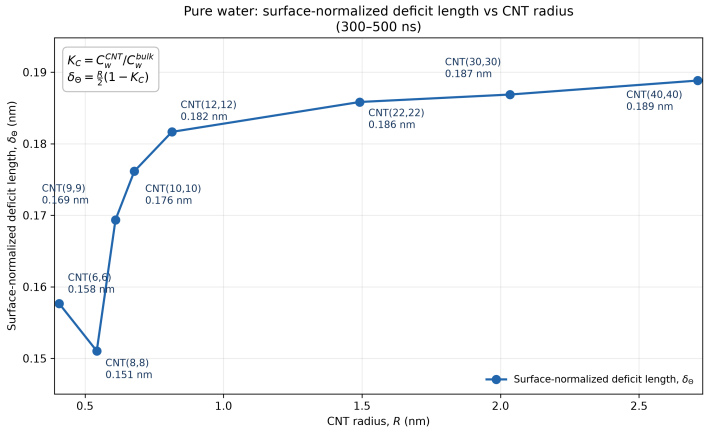


Figure 3: Surface-normalized deficit length versus CNT radius over 300–500 ns.

3.3 Axial Water Diffusion

The MSD is fitted linearly, giving

$$\mathrm{MSD}_z(t) = 2D_z t + b, \qquad D_z = \frac{1}{2}\frac{\mathrm{d}\,\mathrm{MSD}_z(t)}{\mathrm{d}t}. \tag{5}$$

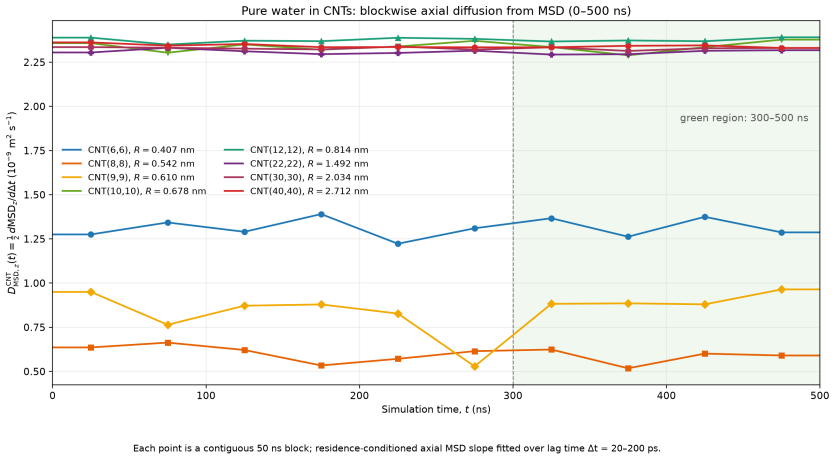


Figure 4: Axial water diffusion over 0–500 ns. Each point is a separate 50 ns block estimate plotted at its midpoint. Each D_z value is obtained from the slope of the axial MSD over $t = 20$–200 ps; the shaded region in Figure 4 marks 300–500 ns.

4 Water and Ion Partitioning

This section examines saltwater behavior in rigid CNTs from three complementary perspectives. Section 4.1 describes the radial organization of confined water together with water and salt occupancy. Section 4.2 compares ion-related concentration-deficit lengths for Na^+ and Cl^-. Section 4.3 summarizes blockwise axial diffusion estimates for water and ions. Together, these results show how CNT size and nominal reservoir NaCl concentration affect composition, structure, and mobility.

4.1 Water Distribution and Salt Occupancy

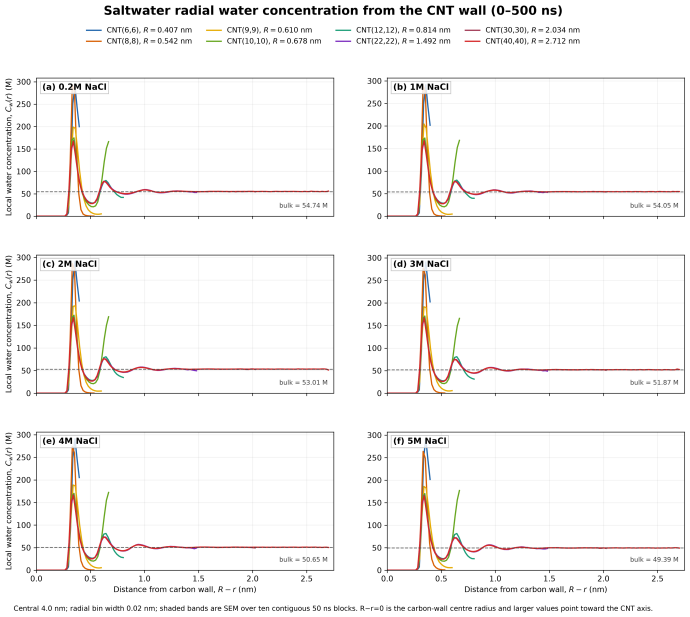


Figure 5: Radial water concentration profiles in the central 4.0 nm of the CNT for six nominal reservoir NaCl concentrations. The dashed line denotes the measured reservoir water concentration; $R-r$ increases from the carbon-wall-centre radius toward the CNT axis.

The near-wall peaks and damped oscillations describe local radial organization, while wider-tube centers approach the reservoir concentration. These local values should be distinguished from the tube-averaged concentration.

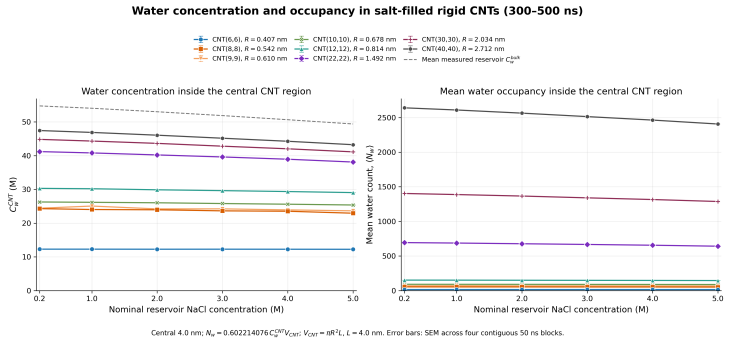


Figure 6: Water concentration and mean water occupancy in the central 4.0 nm CNT region versus nominal reservoir NaCl concentration, 300–500 ns. The dashed reference is obtained from the measured reservoir water concentration.

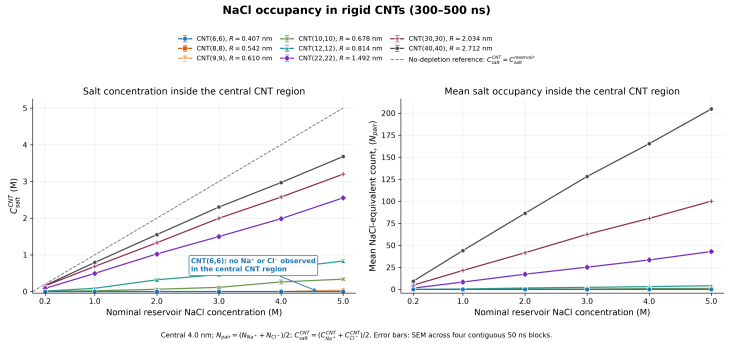


Figure 7: Salt concentration and NaCl-equivalent occupancy in the central CNT region versus nominal reservoir NaCl concentration over 300–500 ns. The dashed line marks the no-depletion reference.

Figures 5–7 show that CNT size and salt concentration affect radial water structure, water occupancy, and salt occupancy. Radial profiles describe local structure, whereas central-region concentrations provide the quantities used for partitioning analysis.

4.2 Surface-Normalized Ion Concentration Deficit

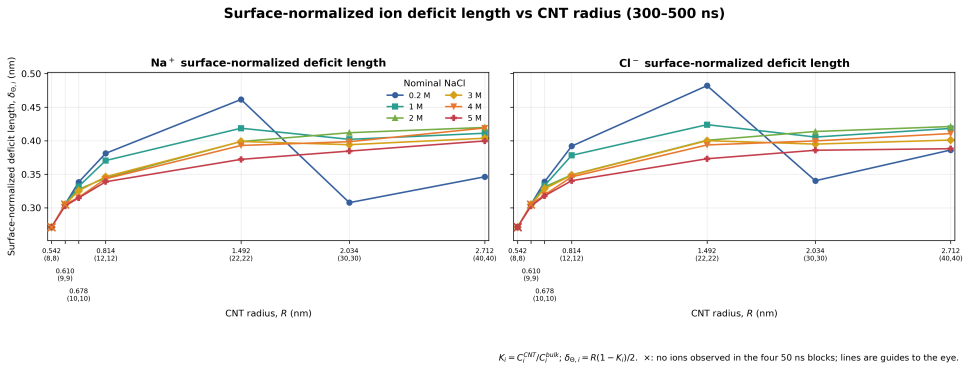


Figure 8: Surface-normalized ion concentration-deficit lengths for Na^+ and Cl^- over 300–500 ns. Crosses indicate conditions with no ions observed in the four 50 ns blocks.

For $i = Na^+$ or Cl^-, $K_i = C_i^{CNT}/C_i^{bulk}$, and the plotted surface-normalized concentration-deficit length is

$$\delta_{\theta,i} = \frac{R}{2}(1 - K_i). \tag{6}$$

Crosses denote unreported estimates caused by insufficient ion observations.

4.3 Axial Water and Ion Diffusion

The $(30, 30)$ and $(40, 40)$ systems are used as representative wide-CNT examples because their larger cross sections provide more consistently reportable water and ion blocks than the narrowest tubes. This selection allows the three species to be compared within a common, readable set of panels while retaining the dependence on nominal salt concentration. Results from the other CNT sizes are provided in Appendix A; missing ion estimates indicate insufficient reporting criteria rather than zero diffusion.

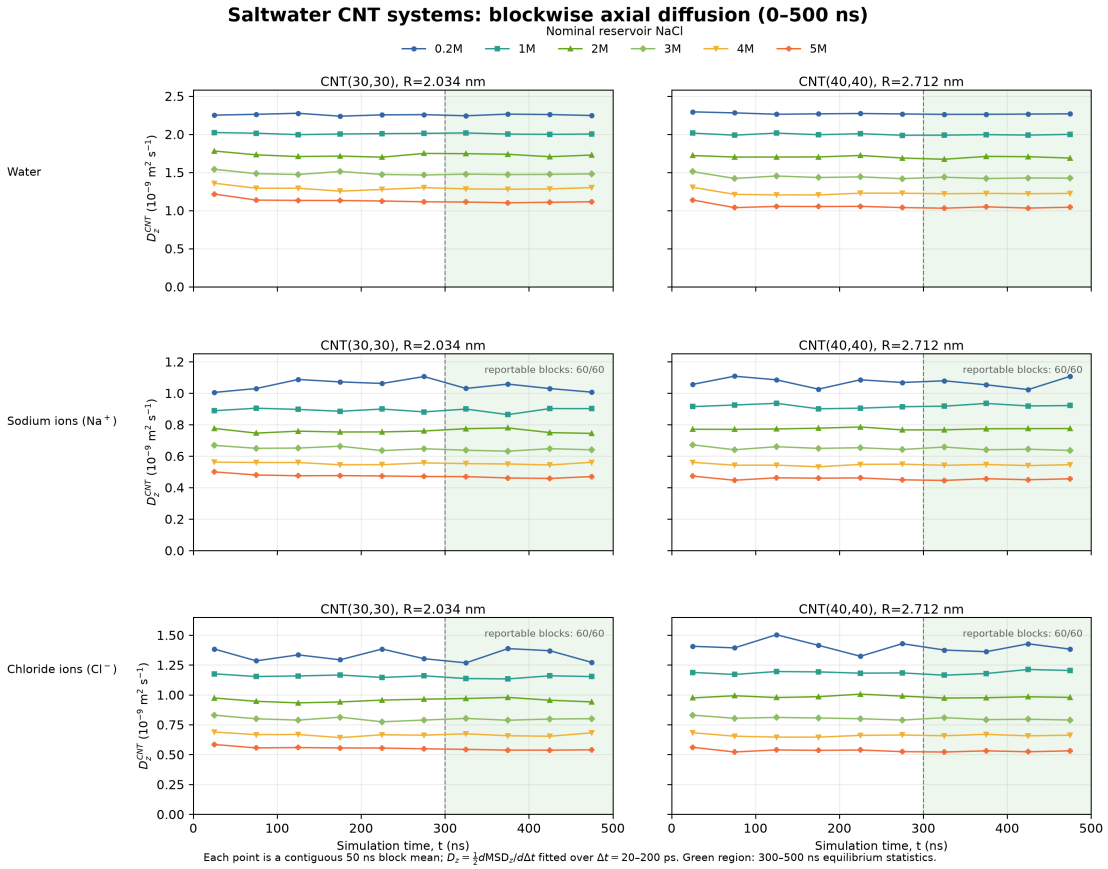


Figure 9: Blockwise residence-conditioned axial diffusion estimates for water, Na^+, and Cl^- across the 0–500 ns trajectories. Each point is obtained from a separate 50 ns block using the axial MSD fit over t = 20–200 ps. Shading marks the 300–500 ns analysis window. Missing ion points indicate unreported estimates and do not represent zero diffusion.

5 Confined Structure

This section focuses on the static structure of confined water and its dependence on CNT size and salt concentration. Static structure factors are used to compare axial and circumferential ordering in pure-water and saltwater systems.

5.1 Static Structure Factor

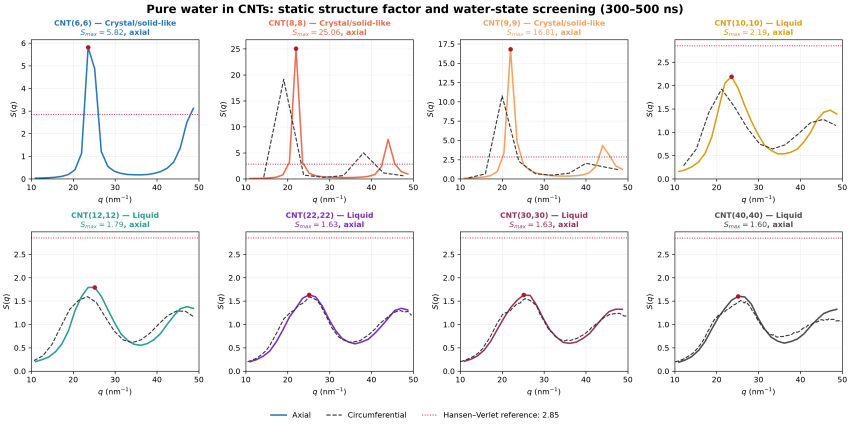


Figure 10: Pure-water static structure factors for the eight CNT sizes over 300–500 ns. The panels show axial and circumferential $S(q)$ curves and the screening threshold used in the analysis.

The static structure factor is defined as

$$S(q) = \frac{1}{N}\left|\sum_{j=1}^{N} e^{i\mathbf{q}\cdot\mathbf{r}_j}\right|^2 . \tag{7}$$

The pure-water curves show pronounced peaks for the narrower CNTs and lower, broader maxima in the wider tubes. Using the adopted $S_{\max}$ threshold, the narrow cases are classified as strongly ordered screening and the wider cases as liquid-like screening. These labels summarize an empirical structural criterion and do not, by themselves, establish a crystalline phase or a thermodynamic transition.

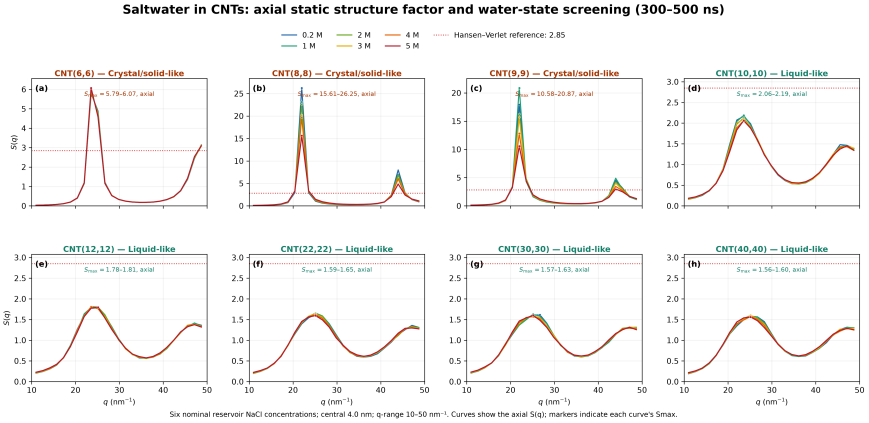


Figure 11: Saltwater axial static structure factors for the eight CNT sizes at six nominal reservoir NaCl concentrations over 300–500 ns. Curves show the concentration dependence of $S(q)$ within each CNT size.

Across the six nominal NaCl concentrations, the saltwater $S(q)$ curves largely preserve the size-dependent pattern observed in pure water. Concentration-dependent changes in peak magnitude are most apparent in the narrower CNTs, whereas the wider CNTs show closely overlapping liquid-like curves. The screening labels are retained for comparison with the pure-water analysis and should not be interpreted as independent phase assignments.

5.2 Structural-State Summary

Table 2: Summary of $S_{\max}$-based structural screening for pure water and six nominal NaCl concentrations across the eight CNT sizes.

CNT	R (nm)	0.2 M	1 M	2 M	3 M	4 M	5 M	Pure $S_{\max}$	Pure-water	Saltwater
(6,6)	0.407	5.793	5.829	5.909	5.962	5.978	6.073	5.815	Strongly ordered	Strongly ordered
(8,8)	0.542	26.246	23.127	22.608	20.208	19.695	15.606	25.065	Strongly ordered	Strongly ordered
(9,9)	0.610	17.899	20.874	16.237	15.706	12.807	10.578	16.806	Strongly ordered	Strongly ordered
(10,10)	0.678	2.166	2.188	2.123	2.097	2.059	2.061	2.193	Liquid-like	Liquid-like
(12,12)	0.814	1.785	1.789	1.814	1.805	1.804	1.798	1.792	Liquid-like	Liquid-like
(22,22)	1.492	1.625	1.654	1.625	1.642	1.617	1.595	1.634	Liquid-like	Liquid-like
(30,30)	2.034	1.614	1.627	1.593	1.593	1.571	1.600	1.631	Liquid-like	Liquid-like
(40,40)	2.712	1.571	1.582	1.598	1.601	1.574	1.562	1.600	Liquid-like	Liquid-like

The screening indicates a transition from strongly ordered behavior in the narrowest CNTs to liquid-like behavior in wider tubes under the adopted $S_{\max}$ criterion. Salt concentration changes the measured $S_{\max}$ values most noticeably in the narrower CNTs, whereas the wider tubes show relatively weak variation. These labels are empirical structural descriptors and do not establish thermodynamic phases.

6 Partitioning and Axial Diffusion Ratios

This chapter focuses on normalized concentration and diffusion ratios. Absolute concentrations, occupancies, absolute D_z, and static structure factors are presented in Sections 3-5 and are not repeated here.

6.1 Pure-Water Ratios

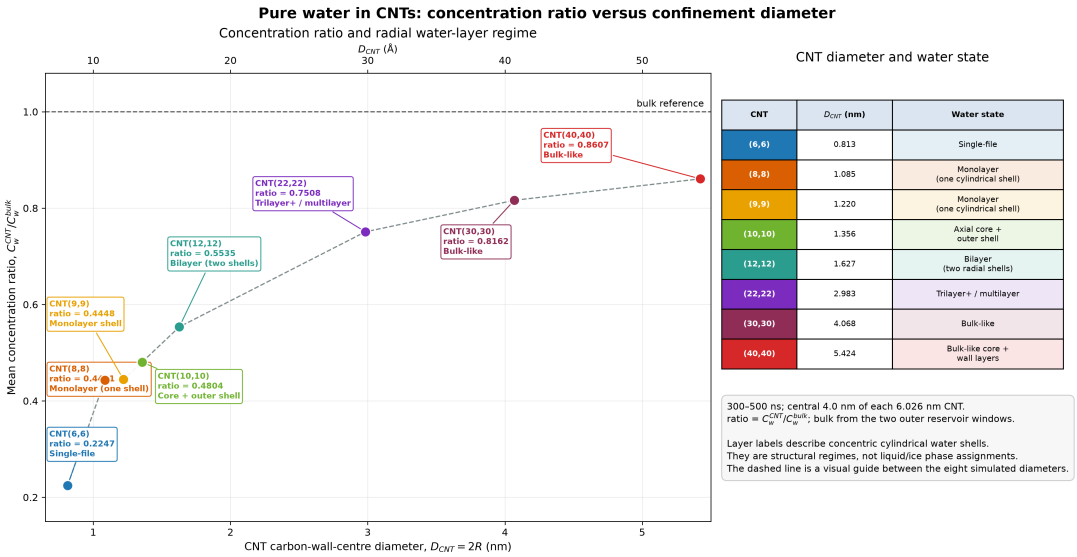
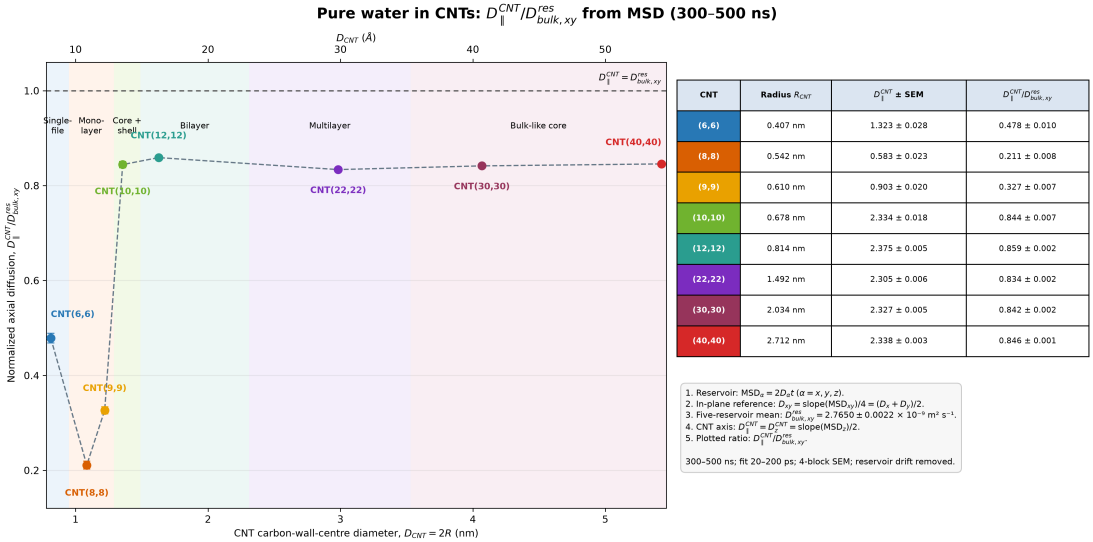


Figure 12: Pure-water concentration ratio $K_C = C_w^{\mathrm{CNT}}/C_w^{\mathrm{bulk}}$ over 300–500 ns as a function of CNT diameter. The ratio compares the central-CNT water concentration with the reference concentration measured in the outer reservoir windows and generally increases with CNT diameter. The embedded table summarizes the corresponding water-state classifications.



CNT	Radius R_{CNT}	$D_{\parallel}^{CNT}$ ± SEM	$D_{\parallel}^{CNT}/D_{bulk,xy}^{res}$
(6,6)	0.407 nm	1.323 ± 0.028	0.478 ± 0.010
(8,8)	0.542 nm	0.583 ± 0.023	0.211 ± 0.008
(9,9)	0.610 nm	0.903 ± 0.020	0.327 ± 0.007
(10,10)	0.678 nm	2.334 ± 0.018	0.844 ± 0.007
(12,12)	0.814 nm	2.375 ± 0.005	0.859 ± 0.002
(22,22)	1.492 nm	2.305 ± 0.006	0.834 ± 0.002
(30,30)	2.034 nm	2.327 ± 0.005	0.842 ± 0.002
(40,40)	2.712 nm	2.338 ± 0.003	0.846 ± 0.001

1. Reservoir: $MSD_\alpha = 2D_\alpha t$ ($\alpha = x, y, z$).
2. In-plane reference: $D_{xy} = \text{slope}(MSD_{xy})/4 = (D_x + D_y)/2$.
3. Five-reservoir mean: $D_{bulk,xy}^{res} = 2.7650 \pm 0.0022 \times 10^{-9}$ m² s⁻¹.
4. CNT axis: $D_{\parallel}^{CNT} = D^{CNT} = \text{slope}(MSD_z)/2$.
5. Plotted ratio: $D_{\parallel}^{CNT}/D_{bulk,xy}^{res}$.

300–500 ns; fit 20–200 ps; 4-block SEM; reservoir drift removed.

Figure 13: Normalized axial self-diffusion coefficient of pure water confined in carbon nanotubes, $D_z^{\mathrm{CNT}}/D_{\mathrm{bulk},xy}^{\mathrm{res}}$, as a function of the CNT carbon-wall-to-wall diameter $D_{\mathrm{CNT}} = 2R$. Diffusion coefficients were obtained from the mean-squared displacement over 300–500 ns and normalized by the in-plane diffusion coefficient of water in the outer reservoir. The results show a strong diameter dependence, with a minimum in the narrow single-file regime and a gradual increase toward the bulk-like regime.

6.2 Saltwater Ratios

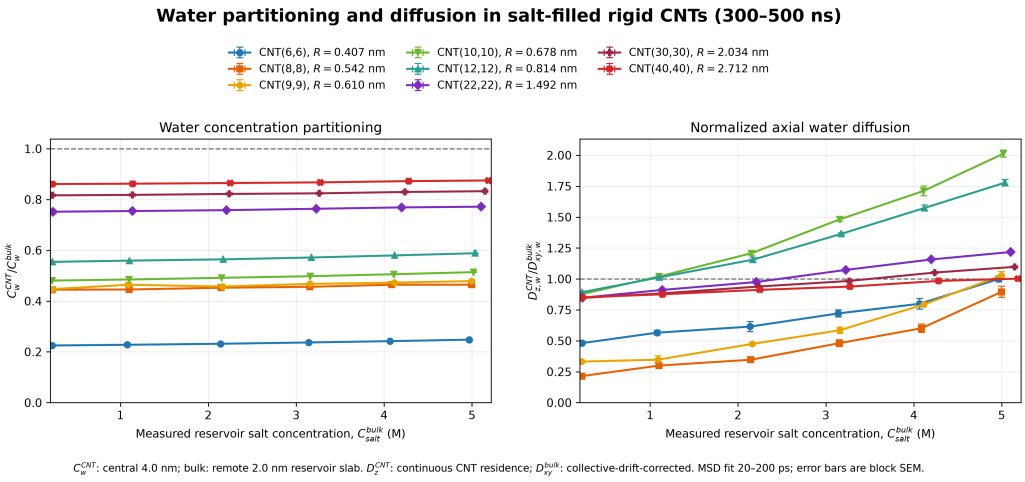


Figure 14: Water partitioning and normalized axial diffusion in salt-filled rigid carbon nanotubes as a function of the measured reservoir salt concentration. The left panel shows the water concentration ratio, $C_w^{\mathrm{CNT}}/C_w^{\mathrm{bulk}}$, while the right panel shows the normalized axial water diffusion coefficient, $D_z^{\mathrm{CNT}}/D_{\mathrm{bulk},xy}^{\mathrm{res}}$, for CNTs of different radii. Data were collected over 300–500 ns. Diffusion coefficients were obtained from axial mean-squared-displacement fits over 20–200 ps.

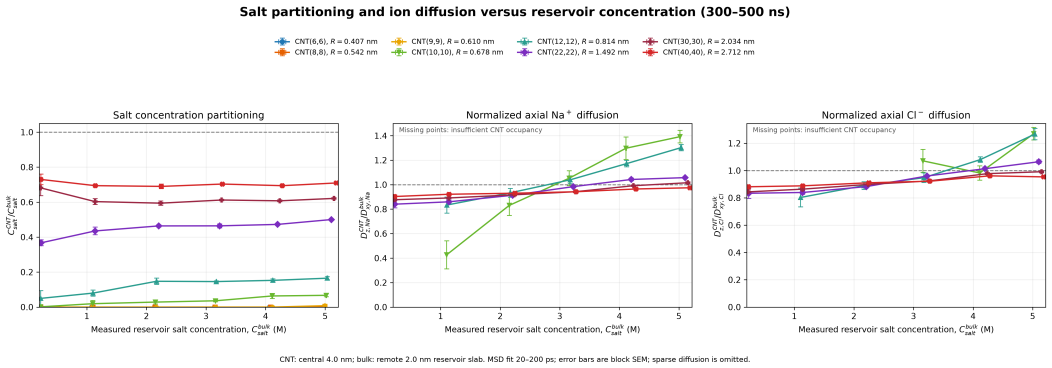


Figure 15: Salt partitioning and ion diffusion in rigid carbon nanotubes as a function of the measured reservoir salt concentration over 300–500 ns. The left panel shows the salt concentration partitioning ratio, $C_{\mathrm{salt}}^{\mathrm{CNT}}/C_{\mathrm{salt}}^{\mathrm{bulk}}$. The middle and right panels show the normalized axial diffusion coefficients of Na^+ and Cl^-, respectively, $D_{z,\mathrm{Na}}^{\mathrm{CNT}}/D_{z,\mathrm{Na}}^{\mathrm{bulk}}$ and $D_{z,\mathrm{Cl}}^{\mathrm{CNT}}/D_{z,\mathrm{Cl}}^{\mathrm{bulk}}$, for CNTs with different radii. The CNT region was defined as the central 4.0 nm, and the bulk reference was obtained from a remote 2.0 nm reservoir slab.

7 Discussion and Limitations

The present interpretation is based on rigid CNTs, finite 500 ns trajectories, and finite sampling for each condition. Ion diffusion in the narrowest CNTs is particularly affected by sparse reportable sampling. The residence-conditioned MSD provides a specified axial-diffusion estimate, but it is not an estimate of a nonequilibrium transport flux. Likewise, the $S_{\max}$ labels are empirical screening criteria rather than thermodynamic phase assignments. Missing or unreported points indicate that the reporting standard was not met and do not represent zero diffusion.

8 Conclusions

The results indicate that CNT size sets the principal structural regime of confined water. Across the investigated geometries, changes in radial organization and static structure-factor screening provide a size-dependent framework for interpreting the confined liquid. The structural labels remain empirical descriptors of the adopted screening procedure.

At a given CNT geometry, changing the nominal salt concentration modifies the composition of the confined region. Water and salt partitioning therefore depend jointly on the available confined volume and the reservoir condition, with measured reservoir concentrations used for the normalized comparisons.

The normalized axial diffusion results show species- and size-dependent responses for water, Na^+, and Cl^-. Their ratios provide a cross-system comparison of mobility while retaining the distinction between reportable estimates and missing ion data. The pure-water and saltwater water responses can therefore be compared without repeating the underlying absolute blockwise diffusion results.

Taken together, partitioning, static structure factors, and normalized diffusion provide complementary views of the same confined systems. CNT size sets the structural regime, salt concentration modifies the composition within that regime, and the resulting partitioning and structural organization are reflected in species-dependent axial diffusion. These relationships are presented as correlations among observables rather than as evidence that a structural change directly causes a diffusion change. These results support a geometry–composition–dynamics framework for confined aqueous systems: CNT size sets the structural regime, salt concentration modifies the composition within that regime, and the resulting partitioning and structural organization are reflected in species-dependent axial diffusion.

A Supplementary Methods and Tables

A.1 Lennard–Jones parameters

The topology uses Lorentz–Berthelot mixing (combination rule 2). The C–OW pair uses the explicit values below, and the C–C nonbonded LJ interaction is explicitly set to zero.

Table 3: Lennard–Jones atom-type parameters and the C–OW pair override.

Type or pair	σ (nm)	ϵ (kJ mol^{-1})
C	0.3568	0.2966
OW	0.3166	0.6500
HW	0	0
Na	0.2310	0.4500
Cl	0.4300	0.4200
C–OW override	0.3367	0.4391

For atom pairs without a specific override, the Lorentz–Berthelot mixing rules are used:

$$\sigma_{ij} = \frac{\sigma_i + \sigma_j}{2}, \qquad \epsilon_{ij} = \sqrt{\epsilon_i \epsilon_j}.$$

The C–OW pair is an exception and uses the explicit parameters:

$$\sigma_{\mathrm{C-OW}} = 0.3367 \text{ nm}, \qquad \epsilon_{\mathrm{C-OW}} = 0.4391 \text{ kJ mol}^{-1}.$$

A.2 Supplementary Partitioning Tables

Table 4: Pure-water surface-normalized deficit-length data.

CNT	R (nm)	K_C	δ_θ (nm)
(6,6)	0.4067	0.224650	0.157663
(8,8)	0.5424	0.443111	0.151034
(9,9)	0.6101	0.444775	0.169371
(10,10)	0.6781	0.480420	0.176156
(12,12)	0.8137	0.553492	0.181657
(22,22)	1.4916	0.750832	0.185834
(30,30)	2.0340	0.816249	0.186875
(40,40)	2.7120	0.860744	0.188830

The water partition coefficient and surface-normalized deficit length are defined as

$$K_C = \frac{C_w^{\mathrm{CNT}}}{C_w^{\mathrm{bulk}}}, \qquad \delta_\theta = \frac{R}{2}\left(1 - K_C\right).$$

Table 5: Surface-normalized Na^+ concentration-deficit lengths.

CNT	R (nm)	0.2 M	1 M	2 M	3 M	4 M	5 M
(8,8)	0.5424	0.27121	0.27121	0.27121	0.27116	0.27121	0.27121
(9,9)	0.6101	0.30505	0.30505	0.30505	0.30262	0.30505	0.30505
(10,10)	0.6781	0.33815	0.33178	0.32745	0.32534	0.31469	0.31469
(12,12)	0.8137	0.38118	0.37042	0.34631	0.34223	0.33882	0.34456
(22,22)	1.4916	0.46143	0.41857	0.39880	0.39880	0.39289	0.37220
(30,30)	2.0340	0.30768	0.40203	0.41190	0.39406	0.39851	0.38453
(40,40)	2.7120	0.34619	0.41111	0.41976	0.40367	0.41894	0.39962

Table 6: Surface-normalized Cl^- concentration-deficit lengths.

CNT	R (nm)	0.2 M	1 M	2 M	3 M	4 M	5 M
(8,8)	0.5424	0.27121	0.27121	0.27121	0.27121	0.27121	0.27121
(9,9)	0.6101	0.30505	0.30505	0.30505	0.30262	0.30505	0.30505
(10,10)	0.6781	0.33895	0.33312	0.32833	0.31942	0.31780	0.31780
(12,12)	0.8137	0.39192	0.37817	0.34865	0.34613	0.34043	0.34912
(22,22)	1.4916	0.48204	0.42390	0.40061	0.40004	0.39370	0.37311
(30,30)	2.0340	0.34032	0.40545	0.41372	0.39493	0.39967	0.38576
(40,40)	2.7120	0.38614	0.41829	0.42114	0.40096	0.41065	0.38807

The ion partition coefficient and the corresponding surface-normalized deficit length are defined as

$$K_i = \frac{C_i^{\mathrm{CNT}}}{C_i^{\mathrm{bulk}}}, \qquad \delta_{\theta,i} = \frac{R}{2}\left(1 - K_i\right),$$

where i denotes either Na^+ or Cl^-.

Table 7: Initial particle counts and initial average salt concentrations calculated from the nominal NaCl composition.

CNT	Nominal NaCl	n_{salt}	n_{w}	$C_{\mathrm{salt,ave}}$ (M)
(6,6)	0.2 M	44	12284	0.198795
(6,6)	1.0 M	215	11942	0.999204
(6,6)	2.0 M	416	11540	2.000693
(6,6)	3.0 M	604	11164	3.002687
(6,6)	4.0 M	779	10814	3.998012
(6,6)	5.0 M	944	10484	4.997329
(8,8)	0.2 M	44	12286	0.198763
(8,8)	1.0 M	215	11944	0.999037
(8,8)	2.0 M	416	11542	2.000347
(8,8)	3.0 M	604	11166	3.002149
(8,8)	4.0 M	779	10816	3.997273
(8,8)	5.0 M	945	10484	5.002623
(9,9)	0.2 M	44	12286	0.198763
(9,9)	1.0 M	215	11944	0.999037
(9,9)	2.0 M	416	11542	2.000347
(9,9)	3.0 M	604	11166	3.002149
(9,9)	4.0 M	779	10816	3.997273
(9,9)	5.0 M	945	10484	5.002623
(10,10)	0.2 M	44	12286	0.198763
(10,10)	1.0 M	215	11944	0.999037
(10,10)	2.0 M	416	11542	2.000347
(10,10)	3.0 M	604	11166	3.002149
(10,10)	4.0 M	779	10816	3.997273
(10,10)	5.0 M	945	10484	5.002623
(12,12)	0.2 M	44	12286	0.198763
(12,12)	1.0 M	215	11944	0.999037
(12,12)	2.0 M	416	11542	2.000347
(12,12)	3.0 M	604	11166	3.002149
(12,12)	4.0 M	779	10816	3.997273
(12,12)	5.0 M	945	10484	5.002623
(22,22)	0.2 M	44	12286	0.198763
(22,22)	1.0 M	215	11944	0.999037
(22,22)	2.0 M	416	11542	2.000347
(22,22)	3.0 M	604	11166	3.002149
(22,22)	4.0 M	779	10816	3.997273
(22,22)	5.0 M	945	10484	5.002623
(30,30)	0.2 M	44	12286	0.198763
(30,30)	1.0 M	215	11944	0.999037
(30,30)	2.0 M	416	11542	2.000347
(30,30)	3.0 M	604	11166	3.002149
(30,30)	4.0 M	779	10816	3.997273
(30,30)	5.0 M	945	10484	5.002623
(40,40)	0.2 M	44	12286	0.198763
(40,40)	1.0 M	215	11944	0.999037
(40,40)	2.0 M	416	11542	2.000347
(40,40)	3.0 M	604	11166	3.002149
(40,40)	4.0 M	779	10816	3.997273
(40,40)	5.0 M	945	10484	5.002623

The initial average salt concentration is

$$C_{\mathrm{salt,ave}} = 55.5\,\frac{n_{\mathrm{salt}}}{n_{\mathrm{w}}}.$$

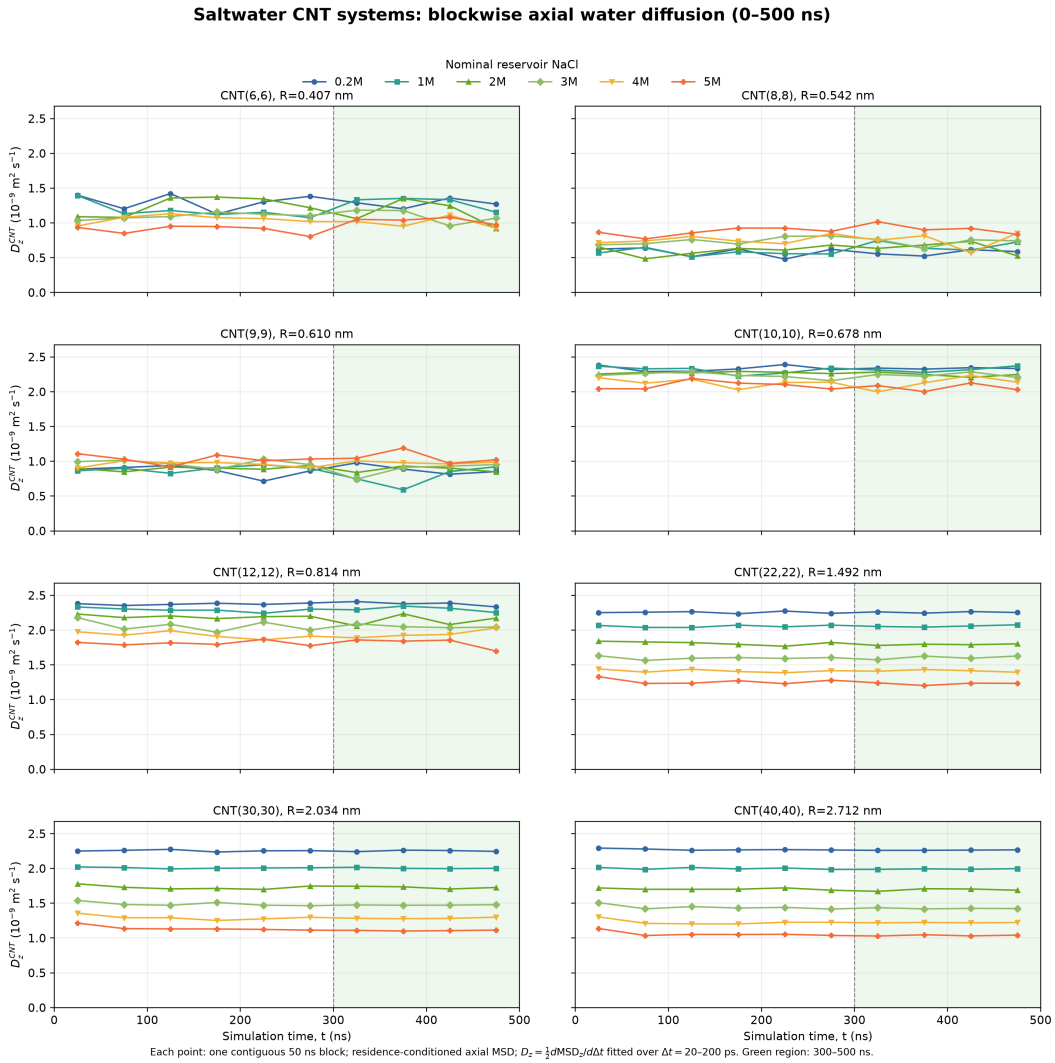


Figure 16: Blockwise water axial diffusion across all eight CNT sizes, 0–500 ns.

Table 8: Supplementary water diffusion data.

CNT	R (nm)	0.2 M	1 M	2 M	3 M	4 M	5 M
(6,6)	0.4067	1.281	1.294	1.146	1.098	1.002	1.037
(8,8)	0.5424	0.571	0.680	0.645	0.725	0.748	0.920
(9,9)	0.6101	0.884	0.778	0.880	0.885	0.984	1.057
(10,10)	0.6781	2.337	2.320	2.247	2.241	2.124	2.062
(12,12)	0.8137	2.378	2.301	2.137	2.055	1.946	1.814
(22,22)	1.4916	2.257	2.058	1.793	1.606	1.414	1.229
(30,30)	2.0340	2.253	2.005	1.729	1.475	1.286	1.108
(40,40)	2.7120	2.264	1.993	1.693	1.426	1.222	1.038

$$D_z = \frac{1}{2}\frac{\mathrm{d}\,\mathrm{MSD}_z(t)}{\mathrm{d}t}$$

$$\left(\text{Unit: } D_{z,w}^{\mathrm{CNT}}\ (10^{-9}\,\mathrm{m^2\,s^{-1}})\right)$$



Figure 17: Blockwise Na^+ axial diffusion across all eight CNT sizes, 0–500 ns. Missing estimates fail the reporting criteria.

Table 9: Supplementary Na^+ diffusion data.

CNT	R (nm)	0.2 M	1 M	2 M	3 M	4 M	5 M
(6,6)	0.4067	–	–	–	–	–	–
(8,8)	0.5424	–	–	–	–	–	–
(9,9)	0.6101	–	–	–	–	–	–
(10,10)	0.6781	–	0.430	0.693	0.721	0.735	0.648
(12,12)	0.8137	–	0.840	0.771	0.711	0.661	0.606
(22,22)	1.4916	0.991	0.862	0.751	0.666	0.581	0.484
(30,30)	2.0340	1.030	0.891	0.762	0.639	0.552	0.465
(40,40)	2.7120	1.066	0.924	0.773	0.644	0.543	0.452

$$D_z = \frac{1}{2}\frac{\mathrm{d}\,\mathrm{MSD}_z(t)}{\mathrm{d}t}$$

$$\left(\text{Unit: } D_{z,Na}^{\mathrm{CNT}}\ (10^{-9}\,\mathrm{m^2\,s^{-1}})\right)$$



Figure 18: Blockwise Cl^- axial diffusion across all eight CNT sizes, 0–500 ns. Missing estimates fail the reporting criteria.

Table 10: Supplementary Cl^- diffusion data.

CNT	R (nm)	0.2 M	1 M	2 M	3 M	4 M	5 M
(6,6)	0.4067	–	–	–	–	–	–
(8,8)	0.5424	–	–	–	–	–	–
(9,9)	0.6101	–	–	–	–	–	–
(10,10)	0.6781	–	–	–	0.932	0.683	0.708
(12,12)	0.8137	–	1.070	0.957	0.819	0.749	0.702
(22,22)	1.4916	1.315	1.115	0.942	0.821	0.692	0.579
(30,30)	2.0340	1.328	1.147	0.961	0.797	0.667	0.540
(40,40)	2.7120	1.388	1.191	0.978	0.796	0.662	0.527

$$D_z = \frac{1}{2}\frac{\mathrm{d}\,\mathrm{MSD}_z(t)}{\mathrm{d}t}$$

$$\left(\text{Unit: } D_{z,Cl}^{\mathrm{CNT}}\ (10^{-9}\,\mathrm{m^2\,s^{-1}})\right)$$